

Jorge Becerra Garrido

Curriculum Vitae¹

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RESEARCH INTERESTS

I am mostly interested in quantum topology. The following is a non-exhaustive list of my interests:

- Quantum knot invariants, especially those which arise from ribbon categories/algebras.
- Tangle categories.
- Finite type invariants and the Kontsevich invariant.
- Modular categories, modular functors, modular operads.
- Factorisation homology, skein categories.

EMPLOYMENT

Université Bourgogne Europe (France). 2024 – present
Postdoc in the research group of L. Woike.

EDUCATION

Ph.D. in Mathematics, *University of Groningen* (The Netherlands). 2019 – 2024
Thesis: *Universal quantum knots invariants*, supervised by R. van der Veen.

M.Sc. Mathematical Sciences, *Utrecht University* (The Netherlands). 2017 – 2019
Honours programme UGC, Cum Laude.
Thesis: *K-theory with Reality*, supervised by L. Meier.

B.Sc. Mathematics, *University of Extremadura* (Spain). 2013 – 2017
Thesis: *Theory of Relativity*, supervised by J.A. Navarro.

PUBLICATIONS & PREPRINTS

1. *Pivotal Module Categories, Factorization Homology and Modular Invariant Modified Traces* (with Lukas Woike). Preprint.
2. *XC-tangles and universal invariants*. Preprint.
3. *Minimal generating sets of rotational Reidemeister moves* (with Kevin van Helden). Preprint.
4. *A refined functorial universal tangle invariant*. Preprint.
5. *Strictification and non-strictification of monoidal categories*. Preprint.
6. *On Bar-Natan - van der Veen's perturbed Gaussians*, Rev. Real Acad. Cienc. Exactas Fis. Nat. Ser. A-Mat., 118(2):46, 2024.
7. *A combinatorial PROP for bialgebras*. European J. Combin. 124 (2025), Paper No. 104086.
8. *An Introduction to Knot Homology Theories*, TEMat monográficos, 2 (2021): Proceedings of the 3rd BYMAT Conference, pp. 11-14. ISSN: 2660-6003.
9. *Finite, Infinite and Twisted Ascent and Descent* (with Jesús MF Castillo), La Gaceta de la Real Sociedad Matemática Española, Vol. 21 (2018), Núm. 3, Págs 497–516.

¹Last updated: 23 December 2025

RESEARCH TALKS

- *XC-algebras and quantum knot invariants*, [Bienal RSME 2026](#), Alicante. Jan 2026
- *Lee homology and the Lee spectral sequence*, [UIC Quantum Topology Seminar](#), Chicago (online). Oct 2025
- *Hopf-algebraic quantum knot invariants*, [GADT groupe de travail](#), Dijon. Nov 2024
- *Universal quantum knot invariants*, [Groningen Quantum Topology Dag](#). May 2024
- *Hopf algebras and knot invariants*, [ICMAT Geometry Seminar](#), Madrid. Jan 2024
- *Drinfeld doubles and Gaussian calculus for universal knot invariants*, [MAPHYAG Seminar](#), Madrid. Jan 2024
- *Quantum invariants of knots and tangles*, [IMUEX Seminar](#), Badajoz. Jan 2024
- *Hopf algebras and the 2-loop polynomial of knots*, [Algebra Seminar](#), Groningen. Dec 2022
- *Three faces of the 2-loop polynomial of knots*, [4th BYMAT](#), online. Nov 2022
- *Universal quantum knot invariants*, [GQT School & Colloquium](#), Zeist. Aug 2022
- *Quantum invariants via perturbed Gaussian expressions*, [Winter Braids XI](#), Dijon Dec 2021
- *A categorical approach to the universal tangle invariant*, [DIAMANT symposium](#), Utrecht. Nov 2021
- *An introduction to knot homology theories*, [3rd BYMAT](#), online. Dec 2020
- *Foundations of stable homotopy theory*, [IMUEX Seminar](#), Badajoz. May 2019
- *Functional inequalities and convergence in diffusion processes*, [13th Meeting on Functional Analysis and Applications](#). Cáceres . Mar 2017

For a list of non-research talks, see sites.google.com/view/becerra/talks.

SELECTED ATTENDANCE

- *Higher Structures: Recent developments and applications*, Hamburg. Sep 2025
- *Matemale spring school*, Matemale. May 2025
- *Algebraic, geometric and quantum topology in Picardie II*, Amiens. Mar 2025
- *Winter Braids XIV*, Bordeaux. Feb 2025
- *Skein days*, Cergy-Pontoise. Dec 2024
- *Algebraic, geometric and quantum topology in Picardie*, Amiens. Mar 2024
- *Swiss Knots 2023*, Regensburg. Sep 2023
- *From Subfactors to Quantum Topology - In memory of Vaughan Jones*, Geneva. Jun 2022
- *Quantum Topology and Geometry (in honor of Vladimir Turaev)*, Paris Jun 2022
- *Winter Braids XI*, Dijon. Dec 2021
- *Georgia Tech Topology Summer School on 4-manifolds*, Online. Jul 2021
- *Monoidal and 2-categories in representation theory and categorification*, Online Dec 2020
- *QUAntum groups, Categorification, Knot invariants, and Soergel bimodules*, Online. Aug 2020

For a complete list, see sites.google.com/view/becerra/attendance.

TEACHING

- **Université Bourgogne Europe:**

Teaching Assistant for *Mathématiques pour Informat./Electron. 2* (Bachelor). Spring 2026

- **University of Groningen:**

Co-lecturer of *Topics in Topology* (Master, universal knot invariants). Spring 2023

Co-lecturer of *Topics in Topology* (Master, Kirby calculus). Spring 2022

Co-lecturer of *Topics in Topology* (Master, grid homology). Spring 2021

Teaching Assistant for *Multivariable Analysis* (Bachelor). Autumn 2020

Co-organiser of the *Knot Theory Seminar* (Bach./Master). Spring 2020

Co-organiser of the *Groningen Topology Seminar* (Bach./Master). Autumn 2019

- **Utrecht University:**

Teaching Assistant for *Topologie en Meetkunde* (Bachelor, π_1). Spring 2019

Teaching Assistant for *Algebraic Topology* (Master, H_n 's and π_n 's). Autumn 2018

Teaching Assistant for *Topologie en Meetkunde* (Bachelor, π_1). Spring 2018

BACHELOR'S THESES SUPERVISION

- Daniël Apol, *Representability of cohomology theories on CW-complexes*, Groningen. 2021

SKILLS

- **Languages:** Spanish (native), English (fluent), French (intermediate), Dutch (basic).
- **Programming languages:**
Python (with [verified certificate](#) by HarvardX), including SAGE, Numpy, Pandas, Matplotlib, NetworkX and Scikit-learn.
Mathematica, C, R-Commander, Fortran, L^AT_EX, Linux.
- **Teaching:** Teacher training certificate.

GRANTS

- [Research Assistant](#), Department of Mathematics, University of Extremadura (Badajoz, Spain). €2,000. 2016–2017
- Fee waiver for talented students, *Summer School on Geometry*, Utrecht (The Netherlands). €400. 2016
- Twelve university course tuition fee waivers for top student of a course. €1,200. 2013–2016
- One-year university tuition fee waiver for top high-school student. €1,000. 2013

ORGANISATION

- Co-organiser of the workshop *Outils algébriques en théorie conforme des champs*. Nov 2025
- Co-organiser of the workshop *Groningen Quantum Topology Dag*. May 2024

PROFESSIONAL ACTIVITIES

- Organiser of a number of seminars and reading groups: the [PhD Seminar](#), the [Knot homology theories reading group](#) or [Quantum Mechanics bookclub](#) at the University of Groningen; and the [Groupe de Travail sur TQFTs \(non-\)semisimples](#) at the Université Bourgogne Europe.
- Speaker at the *European Researchers' Night* and similar outreach activities at University of Extremadura (2014 – 2017).
- Private tutor of English, Mathematics and Physics for high school students (2013 – 2017).